Lesson 23 Pigeonhole Principle (PHP)

$\boxed{\text{PHP}}$ Finite sets A, B, $|A| = n$, $|B| = m$.

If $n > m$, then there are no one-to-one functions from A to B.

A : set of pigeons, B : set of pigeonholes

If we have more pigeons than pigeonholes, then 2 pigeon must share a pigeonhole.

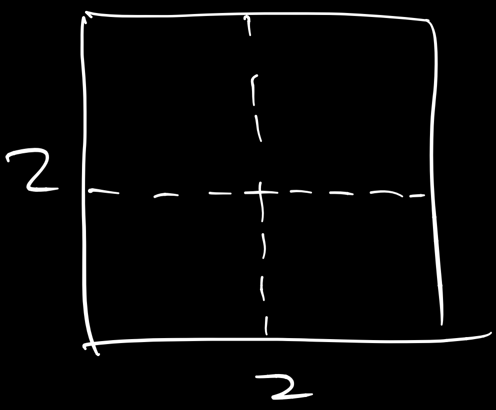
Ex In a class of 30 students, 2 students were born in the same month.

Proof The 30 students are pigeons and the 12 birthmonths are pigeonholes. By PHP, 2 students must share the same birthmonth.

Ex Let $n \in \mathbb{N}$. There are distinct +ve integers a and b s.t. $\underbrace{n^a - n^b}_{n^a \text{ and } n^b \text{ have the same "ones" digit}}$ is $\underbrace{\text{divisible by } 10}_{\text{"ones" (last) digit is } 0}$.

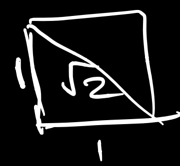
Proof $n^a - n^b$ being divisible by 10 is equivalent to n^a and n^b having the same "ones" digit. Let $n^1, n^2, \ldots, n^{11}$ be the 11 pigeons, and the 10 possible "ones" digits : $0, 1, 2, \ldots, 9$ be the 10 pigeonholes. By PHP, 2 of $n^1, n^2, \ldots, n^{11}$ must have the same "ones" digit.

<u>Ex</u> 5 points in 2×2 square



There are <u>2 points whose distance</u> is at most $\sqrt{2}$.

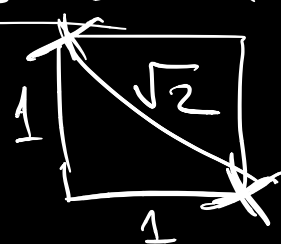
$$\underline{\sqrt{2}} = \sqrt{1^2+1^2}$$



5 points : pigeons

4 unit squares : pigeonholes

By PHP, 2 points <u>must be in the same 1×1 square</u> and they can be at most $\sqrt{2}$ apart.



<u>Generalized PHP</u>

Original PHP : (at least) 2

Generalized PHP : (at least) n

$n \geq 2$

<u>Ex</u> <u>30 students</u>, 12 birthmonths

If there were at most <u>2 students</u> in each month, then there are at most $2(12) = 24$ students

Generalized PHP

If there are n pigeons and m pigeonholes, then there is a pigeonhole with at least $\left\lceil \frac{n}{m} \right\rceil$ pigeons

ceiling

$$\frac{30}{12} = 2.5$$